



TURNKEY INFRASTRUCTURE | CASE STUDY

CCTV & Network Expansion Project

Grasim Cellulosic Division - Vilayat

FIBRE • CAT6 • CCTV • SWITCHING

PROJECT PROFILE

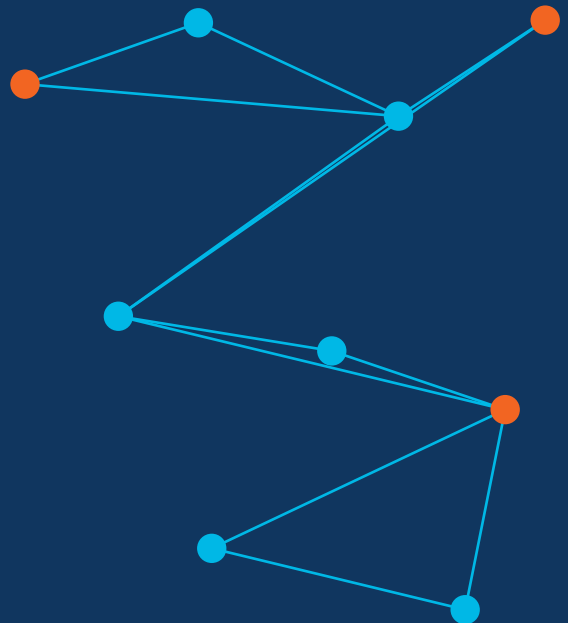
An integrated surveillance and connectivity expansion combining civil route preparation, protected fibre deployment, structured cabling, active networking and camera installation under one accountable execution partner.

TURNKEY DELIVERY

DESIGN COORDINATION • SUPPLY • INSTALLATION • COMMISSIONING

D-TECH SOLUTION INTEGRATORS PVT. LTD.

Lead with Trust. Inspire with Vision.



EXPANSION READY

PROJECT AT A GLANCE

A resilient digital backbone for industrial surveillance

D-TECH was engaged for CCTV and network expansion at Grasim Industries Limited, Grasim Cellulosic Division, Vilayat. The ordered scope connects field cameras to a structured fibre and Cat6 backbone through protected routes, network racks, LIUs and managed switching.

27

Camera installations

2.4 km

Fibre cable route

1.55 km

HDPE protection

1.055 km

Cat6 cabling

Delivery scale

OPTICAL BACKBONE

346 fibre splices supporting a distributed, single-mode optical network.

EDGE CONNECTIVITY

110 UTP termination and commissioning points across the expansion.

FIELD INFRASTRUCTURE

Five intermediate poles, five network racks and 500 m concealed PVC conduit.

ACTIVE NETWORK

Two Aruba Instant On 1930 8G 2SFP switches with fibre interfaces.

D-TECH VALUE PROPOSITION

One execution framework coordinates underground pathways, fibre, copper, network electronics and surveillance endpoints - reducing interface gaps between civil, passive and active workstreams.

INDUSTRIAL EXPANSION REQUIREMENTS

Extending surveillance without fragmenting the network

A CCTV expansion across an operating industrial site requires more than camera mounting. It depends on safe routes, distance-aware media design, protected fibre, clean termination, distributed racks and coordinated commissioning.

DISTANCE & TERRAIN

Camera points distributed across the site required fibre reach, civil excavation and mechanically protected pathways.

MULTI-LAYER DELIVERY

Civil work, passive network, optical splicing, active switching and cameras had to progress as one coordinated project.

INDUSTRIAL SAFETY

Execution needed PPE, approved tools, site induction, risk controls and compliance with Grasim safety requirements.

QUALITY ASSURANCE

Fibre joints, UTP points, racks, LIUs and active components required systematic installation and verification.

PHASED ACCESS

Work fronts were to become available progressively, requiring disciplined resource and schedule coordination.

FUTURE EXPANSION

The infrastructure needed spare capacity and a maintainable topology for subsequent camera or network additions.

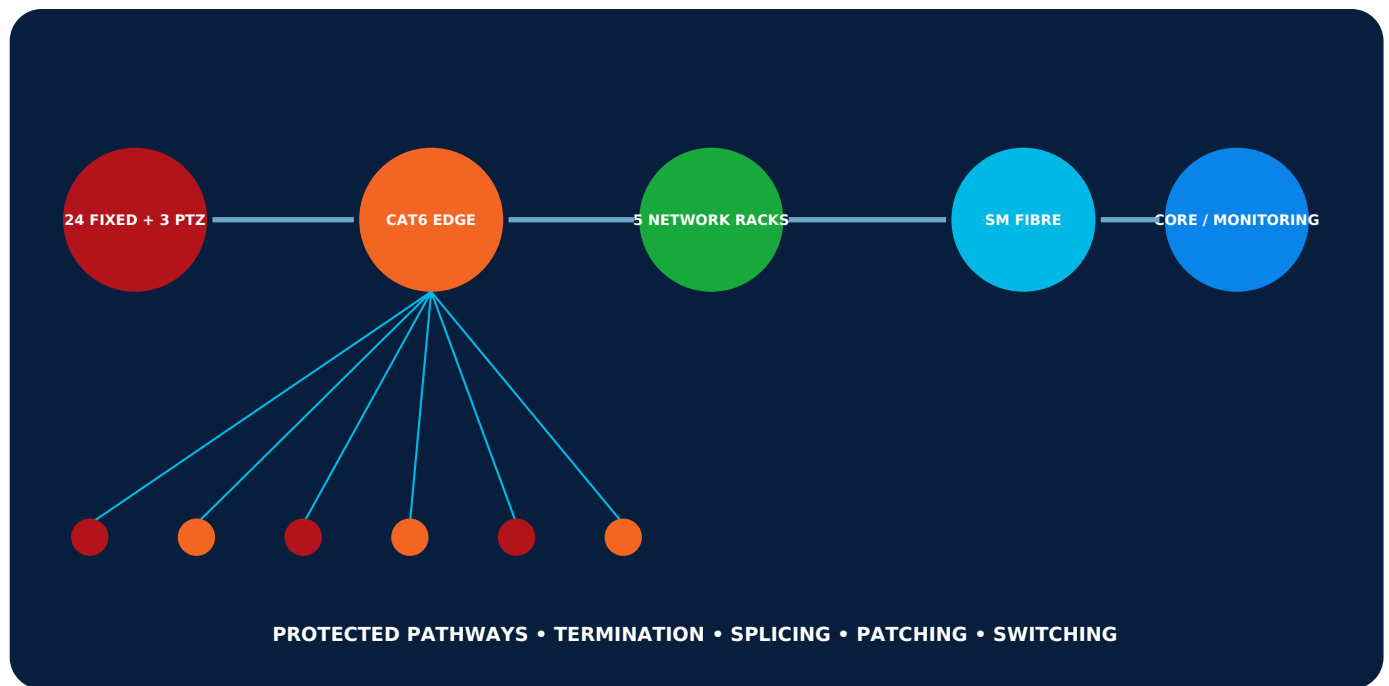
The project objective

Create an expansion-ready surveillance network with reliable field connectivity, protected optical routes and professional commissioning - delivered safely through a single point of accountability.

END-TO-END CONNECTIVITY MODEL

From field cameras to a managed fibre backbone

The solution uses Cat6 for local camera connectivity and single-mode fibre for longer inter-rack links. LIUs, optical patching and Aruba switching create structured hand-off points for operation and maintenance.



Architecture value

REACH

Single-mode fibre supports distributed points beyond practical copper distance.

PROTECTION

HDPE pipe, conduit and disciplined route work protect physical connectivity.

SERVICEABILITY

Racks, LIUs, patch cords and labelled termination points simplify maintenance.

SCALABILITY

Managed edge switching and structured pathways support future additions.

SUPPLY AND SERVICES

A complete installation package

The consolidated scope combines field installation, optical work, copper cabling, pathway construction and active components. Quantities below are taken directly from the supplied Grasim orders.

DELIVERY COMPONENT	VERIFIED QUANTITY
CCTV installation	24 fixed cameras + 3 PTZ cameras
Fibre cable laying	2,400 m
HDPE pipe laying	1,550 m
Cat6 cable laying	1,055 m
Concealed PVC conduit	500 m
Fibre splicing	346 joints
UTP termination / commissioning	110 points
Intermediate poles	5 installations
Network racks up to 15U	5 installations
LIU / fibre patch panels	17 LIU + 4 fibre panel installations
Civil excavation	29 m3

ACTIVE NETWORK SUPPLY

2 x Aruba Instant On 1930 8G 25FP switches, 4 x AMP single-mode OFC patch cords and 2 x AMP 24-port single-mode LIUs with pigtails and couplers.

SAFETY-LED EXECUTION

Eight coordinated delivery points

D-TECH's turnkey method links site readiness, civil work, passive network, active components and commissioning into one controlled delivery sequence.

01 SITE SURVEY

Validate routes, rack points, camera locations, access and work-front readiness.

02 SAFE ROUTES

Excavation, HDPE and conduit deployment with site safety and quality controls.

03 FIBRE LAYING

Install the single-mode optical backbone across protected pathways.

04 SPLICE & TEST

Complete 346 splices, LIU work, patching and optical verification.

05 CAT6 EDGE

Lay Cat6 and commission 110 structured copper endpoints.

06 ACTIVE NETWORK

Install and configure Aruba switching for fibre-enabled connectivity.

07 CCTV DEPLOYMENT

Mount and connect 24 fixed and 3 PTZ camera endpoints.

08 COMMISSIONING

Perform end-to-end checks, rectify punch points and complete handover readiness.

EXECUTION ASSURANCE

The service order requires approved PPE, inspected tools, site induction, risk mitigation, statutory compliance and Grasim safety procedures. Workmanship carries a 12-month on-site warranty from installation and commissioning.

A STRONGER SURVEILLANCE FOUNDATION

Infrastructure designed for reliability and growth

The integrated design strengthens both the surveillance layer and the network that carries it. The benefits below describe the value engineered into the contracted solution; no unverified performance or ROI claim is presented.

END-TO-END ACCOUNTABILITY

One partner coordinates civil pathways, fibre, Cat6, racks, active network and cameras.

RESILIENT CONNECTIVITY

A protected single-mode fibre backbone supports long-distance, industrial-site communication.

MAINTAINABLE DESIGN

Structured racks, LIUs, patching and termination points simplify isolation and future service.

EXPANSION READINESS

Distributed network points and managed switches create a platform for additional cameras.

SAFETY & QUALITY

Execution aligns with Grasim safety procedures, approved PPE and premium passive materials.

LIFECYCLE ASSURANCE

On-site workmanship warranty and Aruba support readiness strengthen post-installation confidence.

D-TECH TURNKEY VALUE PROPOSITION

One network. One surveillance layer. One accountable partner.

A coordinated industrial infrastructure program that connects field execution with professional network engineering, documentation, commissioning and lifecycle support.